



Guidelines Manual	Document Title: Preoperative Medication Management Guidelines	
Originating Entity: MEDICAL Pharmacy Department		
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Date Originated: 7 May, 2020	Revision Due Date: 17th August 2023	Rev. No.: 1
Approved By: Pharmacy and Therapeutic Committee		

I. Introduction

Perioperative period extends from the preoperative day through the operation and into the postoperative recovery period. It is likely that at least 50% of patients undergoing surgery take medications on regular basis. Medical consultants, anesthesiologists, surgeons and pharmacists often must decide if chronic medications should be continued in the perioperative period. Proper perioperative management helps to prevent or minimize complications, to reduce postoperative pain and to accelerate recovery. Many medications should be continued through the perioperative period, with the last dose taken with sips of water two hours prior to the procedure. Only medications known to have perioperative effects or those in common use will be discussed.

II. Principles of medication management:-

The following principles inform the management of chronic medications in the perioperative period:

1-A complete medication history should be obtained, and all clinicians involved in inpatient management should review the medication history.

2-Medication use reported by the patient should be verified to address accuracy of drugs and doses (medication reconciliation).

3-Medication associated with known medical morbidity if withdrawn abruptly should be continued in the perioperative period or tapered if feasible.

4-Intravenous, transdermal, or transmucosal medicines should be substituted when absorption will be impaired because of loss of gastrointestinal function or restriction on oral intake.

5-Medications thought to increase the risk of anesthetic or surgical complications and not essential for the short-term should be held through the perioperative period.

6- Medications not meeting either of the above principles can be



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discontinued or continued based on clinician judgment.

III. Purpose:

To give guidance on the management of preoperative medication at KHCC.

IV. Scope:

This guideline is applicable to KHCC Medical, Pharmacy, and Nursing Departments.

V. Responsibilities:

- 3.1 It is the responsibility of KHCC medical, pharmacy and Nursing departments to follow those guidelines.
- 3.2 Prescribing and administration of per-operative medications at KHCC will be according to the approved indications set by the P&T Committee.

VI. Definitions:

None.

VII. Guidelines:

Categories	Examples	Recommendations	Therapeutic considerations
Cardiovascular medications			
Diuretics	Hydrochlorothiazide Furosemide Spironolactone	<u>Held</u> on the morning of the surgery, except in heart failure patients whose fluid balance is difficult to manage.	-Pay close attention to fluid and electrolytes replacement. -Resume when patient starts taking oral fluids.



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Angiotensin – converting enzyme inhibitors AND angiotensin II receptor blockers (ACEIs and ARBs)	Captopril Enalapril Candesartan Valsartan (NF)	<u>Hold</u> morning dose unless indication is heart failure or poorly controlled hypertension.	- Continuation can result in prolonged hypotension. -Resume ACEIs or ARBs postoperatively as long as the patient is not hypotensive and has normal renal function.
Beta-blockers	Atenolol Bisoprolol Propranolol Metoprolol (NF) Carvidolol	<u>Continue</u> in the perioperative period.	-Reduce ischemia by decreasing myocardial oxygen demand -Help prevent or control arrhythmias -Abrupt withdrawal can lead to hypertension, tachycardia and myocardial ischemia.
Calcium channel blockers	Amlodipine Nifedipine Diltiazem Verapamil	<u>Continue</u> in the perioperative period.	Conflicting evidence on whether there is an increased risk of bleeding.



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Alfa -1 blockers	Doxazosin Tamsulosin Alfuzosin (NF)	<u>Continue</u> in the perioperative period.	May increase the risk of Intraoperative Floppy Iris Syndrome (IFIS) in ophthalmic surgeries.
Central alfa -2 agonists	Clonidine	<u>Continue</u> in the perioperative period.	Abrupt withdrawal can precipitate rebound hypertension and myocardial ischemia.
Non-statin hypolipidemic agents	Niacin (NF) Gemfibrozil Cholestyramine Ezetimibe	<u>Discontinue</u> day before surgery. Resume with oral intake.	They can cause myopathy, rhabdomyolysis and interfere with bowel absorption.
Statins	Simvastatin (NF) Atorvastatin Rosuvastatin (NF)	<u>Continue</u> in the perioperative period.	Continuation may elevate risk of myopathy, but provides cardiovascular protection (reduces inflammation, thrombogenesis and causes plaque stabilization.)



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Digoxin		<u>Continue</u> in the perioperative period.	Decreases the incidence of postoperative supraventricular arrhythmias.
Nitric Oxide Vasodilators	Isosorbide Dinitrate Hydralazine	<u>Continue</u> in the perioperative period.	Hydralazine to be resumed unless blood pressure is low due to risk of hypotension.
Antiarrhythmic	Amiodarone	<u>Continue</u> in the perioperative period	- Drug-drug interactions should be checked. - Suppress arrhythmia
Medications affecting hemostasis			



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Aspirin		The decision to continue or discontinue should depend on: *Bleeding vs. thrombosis risk. *Indication (primary or secondary prophylaxis) *Cardiologist/Surgeon clinical judgement.	
P2Y12 receptor blockers	Clopidogrel Prasugrel (NF) Ticagrelor (NF) Ticopidine (NF)	<u>Discontinue</u> 5-10 days before surgery according to cardiac, bleeding and coagulation risk.	When resuming Clopidogrel consider starting with a loading dose 300-600 mg for 1 day.
Dipyridamole(NF)		<u>Discontinue</u> 2 days before surgery.	No data on its safety if continue in the perioperative period.
Cilostazol (NF)		<u>Discontinue</u> for at least 2-3 days prior to elective surgery.	
Nonsteroidal antiinflammatory drugs (NSAIDs)	Ibuprofen Celecoxib Diclofenac	<u>Discontinue</u> at least 3 days before surgery. Ibuprofen can be discontinued up to 24 hr before elective surgery.	Balance between bleeding risk vs. pain management
Anticoagulants★			



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Vitamin K Antagonists	Warfarin	<u>Discontinue</u> 5 days before surgery.	- Monitor INR the day before surgery, If >1.5 consider low dose oral vitamin K. - Patients with mechanical heart valves,atrial fibrillation and recent thromboembolism may need bridging therapy upon discontinuing or resuming Warfarin ★ ★
Direct thrombin inhibitor	Dabigatran (NF)	<u>Discontinue</u> 2-3 days before surgery.	Bridging therapy may berequired.
Direct factor X inhibitors	Rivaroxaban (NF) Apixaban (NF) Edoxaban (NF) Fondaparinux	<u>Discontinue</u> 2-3 days before surgery. <u>Discontinue</u> Fondaparinux 36-42 hours before surgery	- Resume Fondaparinux6-8 hrs post surgery. - Bridging therapy maybe required.
Low molecular weight heparin (LMWH)	Enoxaparin Tinzaparin	<u>Discontinue</u> 24 hrs before surgery.	Resume 12-24hrs aftersurgery.
Unfractionated heparin (UFH)	Heparin	<u>Discontinue</u> 4-6 hrs before surgery.	Resume 12 hrs aftersurgery.



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★ Assessment for the need of anticoagulation bridging therapy done by cardiologist dependson type of surgery, bleeding vs. thrombosis risk and patient's medical history.

★ ★ Bridging therapy: Start LMWH or UFH therapeutic dose on day -3 prior to surgery. Discontinue LMWH or UFH prior to surgery as above. Resume LMWH and UFH post surgery as above. Start Warfarin 12 hrs post surgery until target INR is reached (2-3 days).

Endocrine agents

Diabetic medications ♦

Sulfonyureas	Glimperide Glibinclamide Gliclazide	<u>Hold</u> morning of surgery.	Increase the risk of hypoglycemia
Biguanides	Metformin	<u>Hold</u> morning of surgery.	Contraindicated in casesof decreased renal perfusion, lactate accumulation or tissue hypoxia.
Thiazolidinediones	Pioglitazone (NF) Rosiglitazone (NF)	<u>Hold</u> morning of surgery.	Worsen fluid retension, peripheral edema and could precipitate congestive heart faluire.
Sodium-glucose cotransporter inhibitors 2	Canagliflozin (NF) Dapagliflozin (NF) Empagliflozin (NF)	<u>Hold</u> morning of surgery.	Increase the risk of hypovolemia.



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Dipeptidyl peptidase-4 And Glucagon-like peptide 2	Sitagliptin (NF) Vildagliptin Saxagliptin (NF) Linagliptin (NF) Teduglutide (NF)	<u>Hold</u> morning of surgery.	May alter Gastrointestinal motility.
Short acting insulin ♦	Aspart (NF) Lispro (NF) Glulisine ((NF)	Must be <u>held</u> while the patient is NPO	Keep Patient on Insulin sliding scale or insulin infusion.
Intermediate acting insulin ♦	Regular insulin (Actrapid)	Must be <u>held</u> while patient is NPO	Keep Patient on Insulin sliding scale or insulin infusion
Long acting insulin ♦	Detemir Glargine Degludec (NF)	<u>Continue</u> in an adjusted dose during the perioperative period.	
♦ Antidiabetic medications should be continued after surgery when the patient resumes oral intake. ♦ For patients who take two types of insulin (intermediate or long and rapid or short-acting) only in the morning, give between <u>one-half to two-thirds</u> of their usual total morning insulin dose (both types of insulin) as intermediate or long-acting insulin to provide basal insulin during the procedure and prevent ketosis. For patients who take insulin (intermediate or long and rapid or short-acting) two or more times per day, give between <u>one-third to one-half</u> of the total morning dose (both types of insulin) as intermediate or long-acting insulin.			
Others			



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Glucocorticoids ■	Prednisone Hydrocortisone Dexamethasone Methylprednisolone	- Nonsuppressed HPA axis: to be <u>continued</u> . - Suppressed HPA axis: to be <u>replaced</u> with a stress dose.	Intermediate/unknown HPA axis suppression: Needs further evaluation; 8AM serum cortisol or ACTH stimulation test.
Selective estrogen receptor modulators (SERMs)	Tamoxifen	For surgeries with moderate to high risk for VTE <u>discontinue</u> 4 weeks prior to surgery when receiving them for breast cancer.	Consultation with oncologist is recommended when SERMs are used for breast cancer treatment.
Levothyroxine		<u>Continue</u> in the perioperative period.	
■ Nonsuppressed HPA axis patients : - Any dose of glucocorticoids for less than 3 weeks. - Morning dose of less than 5mg/day of Prednisone or its equivalent (4mg/day Methylprednisolone, 0.5mg/day Dexamethasone, 20mg/day Hydrocortisone). - Less than 10mg of Prednisone or its equivalent every other day. Suppressed HPA axis patients: - More than 20mg/day of Prednisone or its equivalent for more than 3 weeks. - Any patient on glucocorticoids who has clinical Cushing's syndrome.			
Psychotropic and neurologic agents			



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Tricyclic antidepressants	Amitriptyline	- <u>Continue</u> for patients on high doses without cardiac disease. - Patients on low doses or in whom risk of perioperative arrhythmia is increased <u>taper off</u> over 7 to 14 days prior to surgery increased.	- Resume with oral intake. - Avoid abrupt discontinuation.
Selective serotonin reuptake inhibitors (SSRIs)	Sertraline Citalopram Fluoxetine	- <u>Continue</u> in patients with severe mood disorders. - <u>Discontinue</u> 3 weeks prior to high risk surgeries or in patients requiring ongoing antiplatelet therapy for secondary prevention.	- Resume with oral intake. - Avoid abrupt discontinuation. - Consultation with a psychiatrist is recommended. - Balance between bleeding risk vs psychiatric disorder severity.



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Antipsychotics (Typical and Atypical)	Chlorpromazine Flupentixol (NF) Haloperidol Aripiprazole (NF) Olanzapine Quetiapine (NF) Risperidone	- <u>Continue</u> in the perioperative period in patients with high risk of developing psychoses.	- Avoid abrupt discontinuation. - Consultation with a psychiatrist is recommended. - Balance between QT prolongation risk vs psychiatric disorder severity.
Benzodiazepines	Diazepam Alprazolam Lorazepam	- <u>Continue</u> in the perioperative period.	- Avoid abrupt discontinuation.
Antiepileptic drugs (AEDs)	Phenytoin Phenobarbitone Levetiracetam Carbamazepine Valproic Acid Topiramate Lamotrigine Gabapentin Δ Pregabalin Δ	- <u>Continue</u> in the perioperative period.	Consider serum druglevel monitoring for AEDs with narrow therapeutic window.



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Antiparkinsonian drugs	Carbidopa/Levodopa Pramipexole (NF)	- <u>Hold</u> morning of surgery. - Hold dopamine agonists (Pramipexole) at least 12hrs before surgery.	- Avoid abrupt discontinuation. - Resume with oral inakeas soon as possible post surgery.
Δ Gabapentin and Pregabalin: should be continued in the perioperative period whether used as AEDs or for the management of chronic neuropathic pain in cancer patients.			
Chronic opioid therapy (Opioid dependant patients)			
Methadone		<u>Continue</u> the usual maintenance dose in the perioperative period.	- Has wide interindividual variability due to pharmacodynamic and pharmacokinetic properties: refer to painteam for conversion to IV Morphine when patient is NPO.
			- Can cause QT prolongation/arrhythmias
Morphine sustained release or immediate release		<u>Continue</u> the usual dose in the perioperative period.	- If NPO convert oral dose to IV Morphine; ratio 3:1



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Oxycodone sustained release		<u>Continue</u> the usual dose in the perioperative period.	- If NPO convert oral dose to IV Morphine ☐
Tramadol sustained release or immediate release		<u>Continue</u> the usual dose in the perioperative period.	- If NPO give IV Tramadol as needed.
Fentanyl transdermal patch		- <u>Resumed</u> during the surgery. - In major surgeries, remove patch and administer equipotent dose of oral or IV Morphine ■	- In case patch was removed a new one should be applied ■■
☐ To convert from oral Oxycodone to IV Morphine determine the total daily dose of Oxycodone, calculate the oral Morphine equivalent dose (MED) of Oxycodone (ratio of oral Oxycodone to oral Morphine 1:1.5) then convert the calculated oral MED to IV Morphine (ratio of oral to IV Morphine 3:1). ■ Absorption may be altered according to type of surgery, amount of administered fluids and skin temperature during procedure (hypothermia or rewarming may affect the kinetics of the patch). ■■ Take into consideration that it takes 6 to 12 hours for the new applied Fentanyl patch to re-establish baseline analgesia, so baseline IV Fentanyl infusion may be initiated to maintain plasma concentration.			



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Miscellaneous agents			
Gout therapy	Allopurinol Colchicine	- <u>Continue</u> Allopurinol in the perioperative period. - <u>Hold</u> Colchicine on the morning of the surgery.	Resume Colchicine when patient starts oral intake.
Vascular endothelial growth factor (VEGF)	Bevacizumab	<u>Discontinue</u> at least 6 weeks before surgery	Resume at least 4 weeks after surgery.
Tyrosine kinase inhibitors	Dasatinib Imatinib Sunitinib	<u>Discontinue</u> at least one week before surgery.	- Reinitiate when adequate wound healing occurs. - Affect wound healing and increase bleeding risk.
Antineoplastic agents		Refer to primary oncologist or clinical pharmacist for agent specific information.	- Nadir effect starts 7-14 days after treatment and recovery is usually 2-4 weeks after treatment.
Herbal medications	Ginkgo (NA) Ginseng (NA) St. John's wort (NA)	<u>Discontinue</u> at least one week before surgery.	- Increase bleeding risk, sedative effect of anesthetics and interacts with other medications.

***NF: Non-formulary: Drugs and medical nutrition products which are not currently approved by the P&T Committee for use at KHCC but are registered and available in the U.S.A, United Kingdom, or approved by the EMEA.**



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VIII. Document Historic Track

Version	Section affected	Changes	Approval date
0		Creation of guideline	7 May 2020
1		Reviewed and updated without any change	17th August, 2022